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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : Rakshana . A . R.

Programme: **B.E / B.Tech** IT

Slot : D

Course Code: **CSA1519**

Faculty :Dr.ArulRaj A.M

Course Name: **Cloud Computing and Big Data Analytics**

Date: 3/11/2026

Duration: **1:30 Hr**

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	**Value**
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

CIA - 1:

1) Cloud service Model IaaS, PaaS, SaaS, XaaS.

Introduction:

Cloud computing provides computing resources & software through the internet. Development in hardware, virtualization & networking have made cloud services possible.

a) Developments in Hardware & Internet software:

- Powerful processor improve computing speed.
- Increased RAM & storage support large applications.
- Virtualization allows multiple VMs on one server.
- High speed networks enable faster communication.
- APIs allow different applications to communicate
- Web technologies support browser-based application.

b) Cloud service Models:

1) Infrastructure as a service (IaaS)
- Provides virtual machines, storage & networking
- Users control the operating system & installed software.
- Suitable for streamwave's existing content management application.
- Eg: Amazon EC2.

2) Platform as a service (PaaS).
- Provides tools & runtime environments for application development
- Users need not manage servers or runtime environments.
- Suitable for developing streamwave's new customer-facing application.
- Eg: Google App Engine.

3) Software as a service (SaaS)

- Provides ready-made software through a browser
- No installation or maintenance is required.
- Eg: Salesforce CRM.

4) (XaaS) Everything as a service:

- Provides different IT services through the internet.
- Eg:

Service	Purpose
STaaS	online storage as a service.
DBaaS	Managed Databases as a service
NaaS	Network as a service
SECaaS	Security as a service
Baas	Backup as a service

d) Comparison:

IaaS	Paas	Saas
• Infrastructure as a service	• Development platform as a service	• Ready-made software as a service
• High user control	• Medium control	• Medium control
• User manages the OS	• Provider manages the OS	• Provider manages the OS.

2. Server Consolidation & Energy Saving:

Given:

Original Servers = 15
CPU utilization = 25%
Power per server = 350W
Memory per power = 450W
Target utilization = 65%
Electricity cost = ₹9/KWh
Operating hrs = 8760 hr/year
Fault Tolerance = n+1

a) Active Hosts & Total Hosts:

Total cpu workload:

$$15 \times 25\% = 3.75$$

Required active hosts:

$$N = \frac{3.75}{0.65} = 5.77 \approx 6$$

including N+1:

$$6+1 = 7 \text{ hosts.}$$

b) Consolidation Ratio:

$$\frac{15}{7} = 2.14:1$$

c) CPU utilization:

All 7 hosts active:

$$\frac{3.75}{7} \times 100 = 53.57\%$$

After 1 host fails:

$$\frac{3.75}{6} \times 100 = 62.5\%$$

d) Annual Electricity cost saving:

Original Power:

$15 \times 350 = 5250\ W = 5.25\ KW$

New Power:

$7 \times 450 = 3150\ W = 3.15\ KW$

Annual energy saved:

$(5.25 - 3.15) \times 8766$

$= 18396\ KWh$

Annual cost Saving:

$18396 \times 9 = ₹1,65,564.$

Assumption: All hosts works continously, workload are equal distributed power consumed is normal.

3) CPU & memory over-commitment:

Given:

- Physical CPU cores = 24
- Total RAM = 96GB
- Hypervisor reservation = 8GB
- Number of VMs = 12
- Each VM has 3 VCPUs & 8GB RAM
- Normal CPU usage = 35%, RAM usage = 5GB/VM
- Peak CPU usage = 75%
- Peak RAM = 8 VMs uses 8GB each & 4 VM uses 6GB each.

a) Over commitment Ratio:

1) CPU Ratio:

Total VCPUs:

$12 \times 3 = 36$

$\frac{36}{24} = 1.3 : 1$

2) Memory Ratio:

Available RAM

$96 - 8 = 88$ GB

Allocated RAM:

$12 \times 8 = 96$ GB

$\frac{96}{88} = 1.09 : 1$

b) Average & Peak Memory sufficiency:

Average memory:

$12 \times 5 = 60$ GB

Free memory:

$88 - 60 = 28$ GB.

Peak memory:

$(8 \times 8) + (4 \times 6)$

$= 64 + 24 = 88\,GiB$

Average memory is sufficient. Peak memory exactly use the available RAM, leaving no free capacity.

c) Average & Peak CPU sufficiency:

Average CPU:

$36 \times 35\% = 12.6$

$\frac{12.6}{24} \times 100 = 52.5\%$

Peak CPU: $36 \times 75\% = 27$

$\frac{27}{24} \times 100 = 112.5\%$

Average CPU is sufficient, but peak demand exceeds available CPU capacity by 3 cores.

Issues	Remedies
CPU contention	Add CPU cores & host
Memory shortage	Increase RAM
Poor performance	Monitor resource usage
Unauthorized access	Apply authentication & access control
VM security risks	use VM isolation & regular updates

4. Cloud Federation Between Two Universities:

Introduction:

Cloud Federation connects two independent cloud data centres to share computing resources & services. University can use it to share research, computing facilities & digital library resources.

a) Conventional cloud data-center Architecture:

It consist of
- Servers: Provide computing resources
- Storage: Stores research data & digital books
- Network: Connects both university
- Virtualization: Creates multiple virtual machines
- Resource management: Allocates resources efficiently.

It allows one university to use the available resources of another.

b) Roles of SDDC:
- SD compute: Dynamically allocates CPU & memory
- SD storage: Provides shared storage
- SD Networking: ~~Controls~~ Connect both university security.
- Automation: Monitors & Distributes workload automatically.

c) Role of SOA:
- Uses REST APIs & SOAP for communication
- Allows services to be reused
- Research computing services is supported
- Enables digital-library sharing

Eg: University A researches can access authorized digital library resources hosted by university B

d) Federation challenges & Privacy risks:

Resource sharing – use standard interfaces & tools
service availability – use backup & failover mechanisms
Identity Privacy – implement & uses SSO & identify management
Security threats – Data privacy by Apply encryption
Network latency – use RBAC & access control.

Privacy risk:
- Federated identity information may be exposed
- User presence can reveal online status
- stolen credentials may lead to unauthorized

e) Technical & Administrative safeguards:

Technical:
1) Use MFA & SSO
2) Encrypt data during transmission & storage.
3) Apply role-based access control
4) Maintain audit logs & backups.

Administrative:
1) Establish a federation agreement.
2) Define clat a sharing policies
3) Obtain user consent where required
4) Conduct regular security audit.

5) Edge Filtering, Object Storage & serverless computing:

Given:

- Cameras = 150
- Clip size = 4 MB
- 1 Clip every 3 min
- Storage cost = ₹1.50 per GB/month
- Serverless charge = ₹0.00015 per invocation
- Edge filtering = 25 %
- 1GB = 1000 MB
- Month = 30 days

a) Daily Data volume & Bandwith (without Filtering):

Clips per camera per day:

$$\frac{24 \times 60}{3} = 480$$

Daily data:

$$480 \times 4 \times 150 = 288000 \text{ MB} = 288 \text{ GB}$$

Average bandwith:

$$\frac{288 \times 1000 \times 8}{86400} = 26.67 \text{ Mbps}$$

b) Monthly storage cost (without Filtering):

$$288 \times 30 = 8640 \text{ GB}$$

$$8640 \times 1.50 = ₹12,960$$

c) After Edge Filtering (25 %):

Daily data:

$$288 \times 0.25 = 72 \text{ GB}$$

Average Bandwith:

$$26.67 \times 0.25 = 6.667 \text{ Mbps}$$

Monthly Storage:

$$72 \times 30 = 2160 \text{ GB}.$$

Storage cost:

$2160 \times 1.50 = ₹3,240$

Mothly Saving:

$12960 - 3240 = ₹9,720$

d) Monthly sowerless Invocation cost

Daily clips uploaded:

$72000 \times 0.25 = 18000$

Monthly invocations:

$18000 \times 30 = 540000$

Monthly cost

$18000 \times 0.00015 = ₹81$

Rest API:

It allows cam to communicate with cloud storage using HTTP methods such as POST & PUT to upload video clips.

Why object storage?

- Stores larg video files
- Highly Scalable & durable
- Supports REST APIs acces
- Provides cost effective storage.